Time Series Sales Forecasting

Multi-Model Evaluation & Hybrid Forecasting Framework

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Executive Summary

This project develops and evaluates a structured forecasting framework to predict book sales using historical time series data. The objective was to compare classical statistical methods, machine learning, deep learning, and hybrid approaches to determine which strategy delivers the most reliable forecasts across different demand behaviours.

Using 656 weeks of historical data, models were evaluated on a 32-week out-of-sample horizon. A secondary analysis aggregated weekly data into monthly totals to assess how temporal granularity affects performance.

Key findings:

- Stable seasonal demand can be forecast reliably using SARIMA.
- Nonlinear and volatile demand benefits significantly from hybrid modelling.
- Machine learning does not universally outperform statistical methods.
- Forecast frequency should align with decision-making horizon.
- A flexible modelling framework outperforms a single-model strategy.

This project demonstrates how combining statistical interpretability with machine learning flexibility improves forecasting robustness in real-world scenarios.

1. Business Context

Accurate sales forecasting is critical for:

- Inventory optimisation
- Production planning
- Financial forecasting
- Supply chain efficiency
- Demand risk mitigation

Poor forecasting leads to:

- Overstocking or stockouts
- Lost revenue
- Inefficient capital allocation
- Increased operational risk

The goal of this project was to build a scalable and comparative forecasting framework to support both operational and strategic planning.

2. Data Overview & Preprocessing

The dataset consisted of weekly sales records across multiple ISBNs sourced from four tabs. Each record contained:

- ISBN
- Title
- Week-ending date
- Sales volume

To ensure modelling relevance, ISBNs with sales beyond July 2024 were filtered. A total of 61 active titles met this criterion, preventing distortion from discontinued products.

Two representative books were selected:

- **Book 1:** Exhibits strong annual seasonality and relatively stable demand.
- **Book 2:** Displays higher volatility and irregular sales spikes.

Data was split into:

- Training data: All observations up to 2012-01-01
- Test data: Final 32 weeks

This preserved temporal order and ensured realistic out-of-sample evaluation.

3. Exploratory Demand Behaviour Analysis

Visual inspection revealed distinct structural differences:

Book 1

- Clear 52-week seasonal cycles
- Moderate amplitude fluctuations
- Stable long-term structure

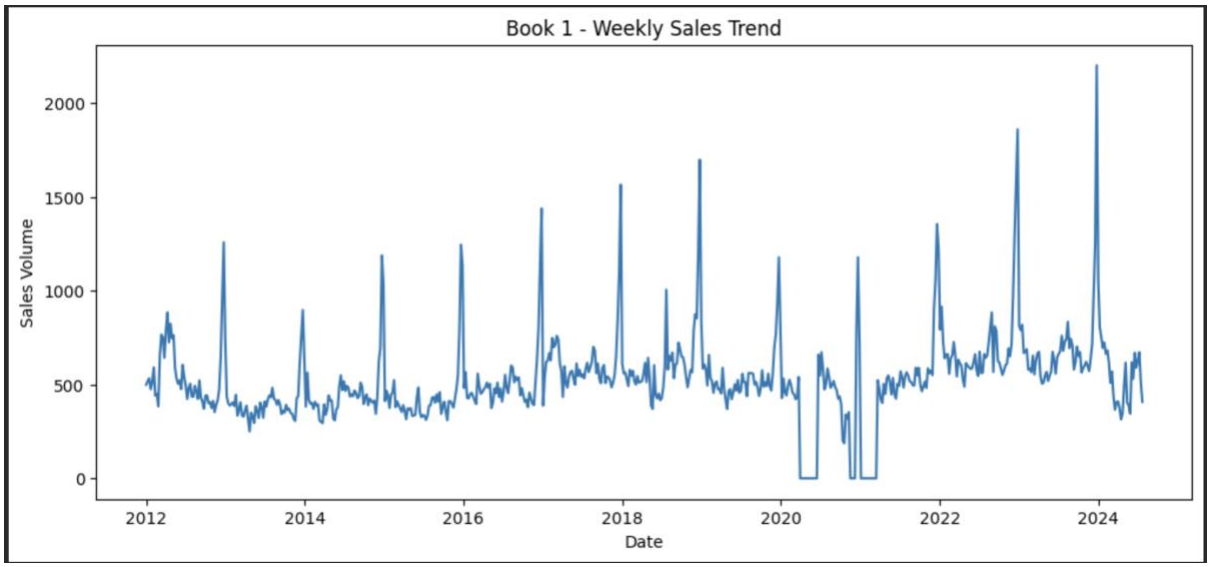

Figure 1: Book 1 - The Alchemist

Book 2

- Higher volatility
- Irregular spikes
- Evidence of nonlinear behaviour

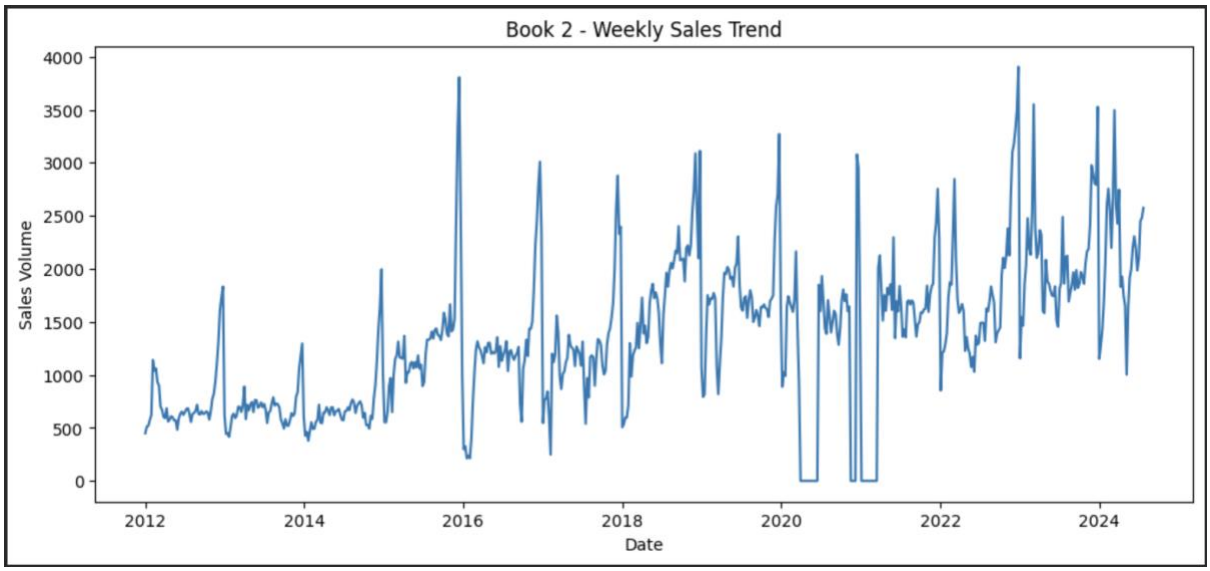

Figure 2: Book 2 - The Very Hungry Caterpillar

This behavioural contrast provided an ideal test case for comparing modelling approaches.

4. Classical Statistical Modelling (SARIMA)

SARIMA models were implemented to capture:

- Trend
- Seasonality (52-week cycle)
- Autoregressive dependencies

Auto ARIMA was used for parameter selection.

Findings:

- Book 1: Seasonal structure captured effectively.
- Book 2: Model captured seasonality but residual analysis indicated unexplained nonlinear components.

Conclusion:
SARIMA performs strongly when demand patterns are stable and seasonally structured but may struggle when nonlinear dynamics dominate.

5. Machine Learning – XGBoost

A supervised learning approach was developed using rolling-window feature engineering. Time-series cross-validation and grid search were applied to optimise hyperparameters.

Findings:

- Book 1: Performance comparable to SARIMA, with limited improvement.
- Book 2: Some improvement in capturing volatility, but not consistently superior.

Key Insight:
Machine learning models require sufficient data volume and meaningful feature structure. When demand patterns are predominantly seasonal and linear, statistical models remain competitive.

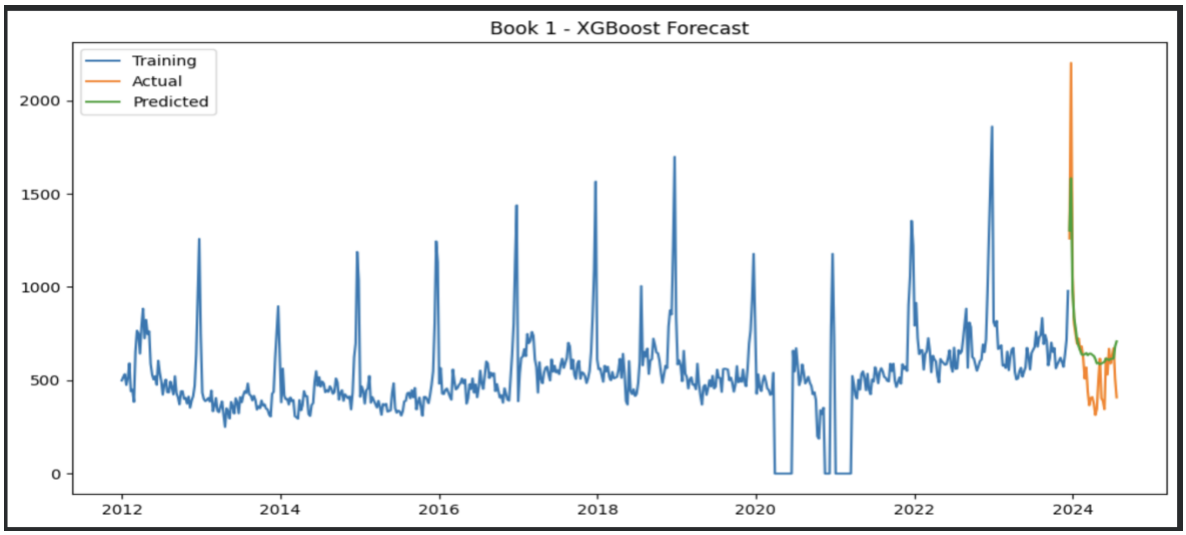

Figure 3: XGBoost Forecast of Book 1

6. Deep Learning – LSTM

LSTM networks were trained to capture nonlinear temporal dependencies. Hyperparameters were tuned using KerasTuner.

Findings:

- Book 1: Moderate performance.
- Book 2: Higher error compared to hybrid approach.

While LSTM provides flexibility, standalone deep learning does not automatically guarantee superior forecasting performance, particularly with moderate dataset sizes.

7. Hybrid Modelling Approaches

Two hybrid strategies were evaluated.

7.1 Sequential Hybrid (SARIMA + LSTM on Residuals)

This approach:

1. Models linear and seasonal components using SARIMA.
2. Trains LSTM on residuals to capture nonlinear structure.
3. Combines forecasts additively.

Results:

- Book 1: Marginal improvement over SARIMA.
- Book 2: Significant reduction in forecast error.

This confirms that residual modelling effectively captures nonlinear behaviour that statistical models alone cannot.

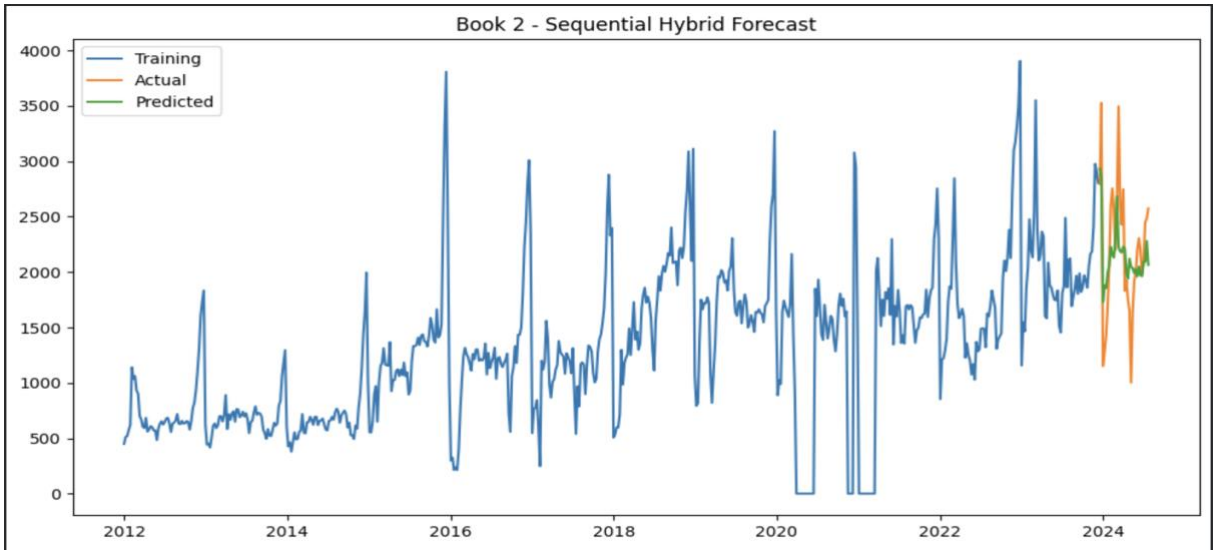

Figure 4: Best Performing Model for Book 2

7.2 Parallel Hybrid (Weighted Combination)

SARIMA and LSTM forecasts were combined using optimised weights.
Results showed that optimal weight heavily favoured SARIMA, indicating that simple averaging is less effective than residual modelling.

Conclusion:
Sequential hybrid modelling is more structurally meaningful than weighted averaging.

8. Monthly Forecasting Analysis

Weekly data was aggregated into monthly totals and forecasted over an 8-month horizon.

Observations:

- Monthly SARIMA outperformed monthly XGBoost.
- Aggregation reduced volatility and strengthened seasonal patterns.
- Machine learning models were impacted by reduced dataset size.

Monthly forecasts were smoother but less responsive to short-term variation.

9. Weekly vs Monthly Comparison

Weekly forecasting:

- Captures short-term operational variation.
- Enables hybrid advantages.
- Better suited for inventory planning.

Table 1: Weekly Performance Summary

Model	Book	MAE
SARIMA	Book 1	141.46
	Book 2	357.42
XGBOOST	Book 1	135.59
	Book 2	464.97
LSTM	Book 1	178.18
	Book 2	832.70
Sequential Hybrid	Book 1	146.06
	Book 2	353.01
Parallel Hybrid	Book 1	155.91
	Book 2	482.41

Monthly forecasting:

- Produces smoother projections.
- Better suited for financial and strategic planning.
- Favours classical statistical models.

Table 2: Monthly Performance Summary

Book	Model	MAE
Book 1	XGBoost	896.74
	SARIMA	773.11
Book 2	XGBoost	2297.60
	SARIMA	1979.89

Forecast granularity materially affects model performance and business applicability.

10. Model Performance Summary

Overall observations:

- SARIMA remains highly competitive for stable demand.
- XGBoost provides incremental gains under moderate volatility.
- LSTM alone does not consistently outperform classical methods.
- Sequential hybrid delivers strongest performance for nonlinear demand.

The results demonstrate that model selection should be demand-driven rather than complexity-driven.

11. Key Business Insights

1. Stable demand patterns do not require deep learning complexity.
2. Nonlinear residual structure justifies hybrid modelling.
3. Forecast frequency should match decision-making horizon.
4. Aggregation changes modelling dynamics.
5. A structured model evaluation framework improves decision confidence.

12. Strategic Recommendations

- Deploy SARIMA for stable seasonal titles.
- Implement sequential hybrid models for volatile or promotion-driven products.
- Use weekly forecasts for operational decisions.
- Use monthly forecasts for budgeting and revenue projections.
- Continuously monitor performance using MAE and MAPE.
- Periodically retrain models to capture structural demand changes.

13. Limitations

- Limited external features (promotions, marketing spend, pricing).
- Moderate dataset size may limit deep learning effectiveness.
- No exogenous variable modelling (SARIMAX not implemented).
- Real-world deployment would require automated retraining pipelines.

14. Future Improvements

- Incorporate external drivers (price, campaigns, holidays).
- Explore Prophet or Gradient Boosting alternatives.
- Evaluate probabilistic forecasting for uncertainty intervals.
- Implement automated model selection framework.
- Deploy as a scalable forecasting pipeline.

Final Conclusion

This analysis confirms that forecasting performance is highly dependent on demand structure and modelling strategy. While classical statistical models remain robust and reliable for stable seasonal products, hybrid approaches significantly enhance accuracy when demand exhibits nonlinear behaviour.

Weekly modelling provides greater responsiveness for operational decision-making, whereas monthly aggregation supports higher-level financial planning. A one-size-fits-all forecasting approach is therefore not optimal.

By implementing a flexible, model-driven forecasting framework aligned with product behaviour and business objectives, the organisation can improve forecast reliability, optimise inventory planning, and reduce demand-related risk.